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PAPER

Objective Mismatch in Federated Learning under Missing Class Support

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SUMMARY Federated learning (FL) often assumes that all classes are observed across clients. However, in many real-world applications such as fall detection, a large fraction of clients contain no samples of rare but critical events. We show that this leads to a fundamental failure mode, where FL appears to converge while remaining biased on rare classes. We theoretically prove that this induces an irreducible bias that cannot be eliminated by longer training or local reweighting. To mitigate this issue, we propose a simple server-side correction and demonstrate its effectiveness on both synthetic data and a real-world dataset.
key words: *federated learning, missing class, stationary bias, convergence analysis, objective mismatch, rare event*

1. Introduction

Federated learning (FL) enables distributed clients to collaboratively train a shared model without exchanging raw data [1], [2]. A central practical obstacle is data heterogeneity: local distributions can differ substantially across clients, causing the FL surrogate objective $F_{\mathrm{FL}}(w) = \sum_k \pi_k F_k(w)$ to diverge from the true global objective $F(w)$. The FL theory community has devoted considerable effort to bounding this divergence—as client drift, gradient dissimilarity, and objective inconsistency—yet all existing analyses share a tacit assumption: every class of interest appears in at least some client's local data. This assumption fails in a large and practically important class of deployments.

Motivating scenario: federated fall detection. Wearable-sensor fall-detection systems for ambient assisted living are an archetypal FL application [3], [4]. Falls are safety-critical events, yet they are extremely rare in practice: epidemiological studies report that most community-dwelling adults fall fewer than twice per year, and many not at all [4]. A fall-detection system federated across 50 smart home gateways therefore faces a situation where, during any given training window, 30 to 40 devices have recorded zero fall events. The *fraction of clients without a single fall sample*—what we call the *missing mass* ρ—can routinely reach 60% to 80%. The same regime arises in industrial anomaly detection, rare-disease monitoring, fraud detection, and environmental event sensing: any domain in which a safety-critical label is temporally sparse [5]. The rare-class prevalence π^* itself spans a wide range depending on data-collection conditions. In free-living, spontaneous-deployment regimes (e.g., continuous fall monitoring of community-dwelling adults), π^* may be on the order of 10^{-2} or smaller. In *moderate rare-event regimes*—typical of curated training corpora and continuous-monitoring scenarios in which event-bearing windows are deliberately over-represented—π^* commonly reaches the 0.1–0.4 range. The theoretical phenomena studied below (missing-class structural mismatch, irreducible bias floor, misleading certificate) hold across the entire range; our experiments evaluate the *moderate* regime ($\pi^* \approx 0.34$ on SisFall) where the effects are already observable on a manageable scale.

Why this is not a class-imbalance problem. A rich line of FL research addresses heterogeneous label distributions via local reweighting, logit calibration, and heterogeneity-aware sampling [6]–[10]. These methods implicitly require that rare-class samples exist somewhere in the client's local data, even if underrepresented. When rare-class support is *entirely absent* from a client, its gradient $\nabla F_k(w)$ carries zero information about the rare-class loss: no scalar reweighting $\alpha_k > 0$ can recover the missing rare-class gradient $\nabla F^{(*)}(w)$ from a client that has never observed class c^*. We formalize this for scalar reweighting in Proposition 3; the same fundamental obstacle—any function purely of c^*-free local data is independent of $\nabla F^{(*)}$—applies structurally to non-scalar local interventions (e.g., logit calibration, restricted softmax), as discussed in Section 2. The failure mode is therefore a *support* problem, not a *frequency* problem—a qualitatively different and more fundamental obstacle.

The hidden certificate problem. A subtlety that makes MC-FL dangerous in practice is that standard FedAvg convergence monitoring based on surrogate-side quantities provides no warning. We prove (Theorem 6) that the surrogate gradient $\|\nabla F_{\mathrm{FL}}(w^{(r)})\|$ continues to decrease to zero as training proceeds, even as the true-objective gradient $\|\nabla F(w^{(r)})\|$ remains permanently elevated above an irreducible floor. A practitioner monitoring training loss or FL stationarity—both quantities derived from the

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surrogate side— would declare convergence and terminate training, unaware that fall-detection recall is silently collapsing. We call this the *misleading certificate* phenomenon (scoped specifically to surrogate-gradient and surrogate-loss monitoring; direct rare-class evaluation on a held-out validation set is, of course, unaffected) and provide explicit conditions under which it occurs.

The theoretical gap. Existing FL convergence analyses (FedProx [11], SCAFFOLD [12], FedNova [13]) bound the gap between client and global objectives under partial participation and heterogeneity, but they assume all classes are represented across the federation as a whole. No prior work has characterized the bias floor incurred when a *fraction* ρ of clients entirely lacks the rare class, shown that this floor is irreducible, or provided closed-form expressions for its dependence on ρ, π^*, and g^*. We fill this gap with a complete theory.

Contributions.

- **MC-FL formalization (Section 3).** We show that the FL surrogate decomposes as $F_{\mathrm{FL}} = (1-\rho)F + \rho F^{(\neg)}$, a convex combination of the true objective and the non-rare risk. This decomposition makes the structural mismatch transparent and shows why no local correction can recover the missing gradient signal.
- **Stationary bias lower bound (Theorem 5).** Any FedAvg-stationary point $\widehat{w}$ satisfies $\|\nabla F(\widehat{w})\| \geq \rho\pi^* g^* / (1-(1-\rho)\pi^*)$, a closed-form floor that grows with both ρ and π^* and cannot be driven to zero by any choice of hyperparameters.
- **Irreducible convergence floor (Theorem 6).** The time-averaged squared gradient satisfies $R^{-1}\sum_r \mathbb{E}\|\nabla F\|^2 \leq \mathrm{UB}_c(R) + \mathrm{UB}_b$, where $\mathrm{UB}_c(R) \to 0$ but $\mathrm{UB}_b = 2\rho^2 B^2/(1-\rho)^2 > 0$ is independent of R, η, E, and K.
- **δ-relaxed theory and monitoring rule (Theorems 8, 9, Corollary 10).** We extend the analysis to approximate missing-class conditions, derive the bias-vanishing boundary $\rho_{\min}(\delta)$, and provide a two-step monitoring protocol.
- **SC-FedAvg (Proposition 11; proof-of-concept).** As a constructive proof that the bias floor is *not fundamental*—i.e., that the floor can be lowered given even minimal server-side rare-class information—we propose SC-FedAvg, a one-line server-side correction. Under a one-step linear approximation, the effective missing mass satisfies $\tilde{\rho} \leq \rho(1-\alpha) < \rho$, strictly lowering the floor; even a single server-held rare-class sample yields $\alpha > 0$. We position SC-FedAvg as evidence that floor-reducing corrections exist, not as a fully established method: full non-asymptotic multi-round guarantees are left for future work.

Section 2 reviews related work. Section 3 develops the theory. Section 4 reports experiments. Section 5 concludes.

2. Related Work

Federated learning foundations. FedAvg [1] introduced local SGD with weight averaging, enabling communication-efficient training without raw-data sharing. Kairouz et al. [2] survey open problems in FL—privacy, robustness, fairness, and heterogeneity—and identify non-IID data as a central challenge, but do not treat the case where an entire class is absent from many clients.

Objective inconsistency and client drift. FedProx [11] adds a proximal term to bound client drift and provides convergence guarantees under heterogeneous distributions and partial participation. SCAFFOLD [12] corrects drift via control variates, achieving faster convergence under non-IID conditions. FedNova [13] identifies an *objective inconsistency* problem arising from heterogeneous local update counts: different clients effectively minimize different surrogates of the global objective. Our work identifies a different and more severe inconsistency: the FL surrogate structurally excludes the rare-class gradient entirely (Lemma 2), an effect that FedNova's re-normalization cannot address.

Label skew and class imbalance in FL. A large body of work tackles heterogeneous label distributions in FL. Wang et al. [6] address class imbalance via local reweighting and oversampling. FedRS [14] restricts the softmax to known local classes to mitigate label-distribution shift. FedLC [7] applies logit calibration based on estimated global label frequencies to correct local update bias. Fed-CBS [8] reduces class imbalance through heterogeneity-aware client sampling. FedBR [9] reduces local learning bias by aligning local feature distributions with a global reference. FedMGD [10] models the global label distribution at the server to guide local optimization. Model concatenation [15] and FedLoGe [16] further exploit label-skew structure through architectural and optimization co-design. We position our result against this body of work in two layers. *(i) Scalar reweighting:* Proposition 3 formally precludes any scalar-reweighted aggregation (e.g., per-client class weighting [6]) from recovering $\nabla F^{(*)}$ when support is zero on a client. *(ii) Non-scalar local interventions:* FedLC [7] (logit calibration), FedRS [14] (restricted softmax), FedBR [9] (feature alignment), and FedMGD [10] (server-modeled global distribution) lie outside the strict scope of Proposition 3, but face the same obstacle *structurally*: for any $k \in \mathcal{M}^*$, every quantity computed from local c^*-free data is independent of $P(\cdot\,|y = c^*)$, so no purely local function can supply the $\nabla F^{(*)}$ direction. External signals such as server priors (FedMGD), known global frequencies (FedLC), or design-time class enumeration

(FedRS) compensate for skew but cannot synthesize the missing direction without additional samples. We treat the rigorous extension of Proposition 3 to the full non-scalar case as a conjecture supported by this structural argument and verified empirically on SisFall (Section 4.4), where LocalReweight shows no improvement over plain FedAvg.MC-FL is therefore not a label-skew problem but a *support problem*: the corrective gradient signal is absent, not merely weak.

Safety-critical and rare-event FL. Liu et al. [3] survey wearable-sensor systems for fall detection and document the extreme sparsity of fall events as a fundamental barrier to learning. Li et al. [4] provide a decade-long review of wearable sensors for fall detection, further confirming that most deployment scenarios yield devices with zero fall events during the observation window. Saad et al. [5] survey FL for anomaly detection and identify rare event scarcity as a key unsolved challenge. These works motivate the MC-FL setting empirically; to our knowledge, no prior theoretical work characterizes FL convergence when class support is entirely absent from a fraction of clients.

Connections to PU learning and label-shift domain adaptation. MC-FL is related to but distinct from two adjacent learning paradigms. *Positive-Unlabeled (PU) learning* [17], [18] trains a binary classifier from positive examples and unlabeled data; methods such as unbiased PU risk estimation rely on a centrally accessible unlabeled pool and a known class prior. In MC-FL, by contrast, the rare-class samples are not absent globally—they are concentrated on a $(1-\rho)$-fraction of clients—and data is locally siloed under FL privacy constraints, so PU estimators are not directly applicable. *Label-shift domain adaptation* [19], [20] treats the case in which the marginal $P(y)$ varies across domains while the class-conditional $P(x|y)$ is preserved. MC-FL is the limit case in which $P_k(y = c^*) = 0$ on a ρ-fraction of domains. Standard label-shift methods estimate domain priors and reweight predictions accordingly, but when $P_k(y = c^*) \equiv 0$, prior estimation is degenerate (estimated prior is exactly zero) and reweighting cannot recover the rare-class direction on those clients. The structural conclusion of Proposition 3 is consistent with both paradigms' fundamental requirement of *some* rare-class signal somewhere; MC-FL identifies a specific federated regime in which this requirement fails on a quantifiable fraction of the federation.

Positioning. In all prior FL theory and label-skew methods, rare-class samples exist somewhere in the federation. In MC-FL they are absent from a fraction ρ of clients, turning a convergence-rate issue into an objective-mismatch issue with an irreducible bias that no amount of training eliminates.

3. MC-FL: Formulation and Analysis

Roadmap. We organize the analysis so that the δ-relaxed bounds (Theorems 8 and 9) are the *practical main results*, and the strict missing-class case (Theorems 5 and 6, under Assumption 1) is the cleanest *special case at* $\delta = 0$, which we present first as an analytically transparent building block. The mathematical content is unchanged from a single combined statement; the strict and relaxed forms differ only in whether the residual term UB_δ (resp. the δ subtraction in the stationary bound) is included. This structure parallels the conventional "ideal then perturbed" presentation in optimization theory and is intended to clarify that all main conclusions of MC-FL persist under approximate missing-class conditions ($\delta > 0$) whenever $\rho > \rho_{\min}(\delta)$.

3.1 Setup

Consider K clients with data sizes n_k and weights $\pi_k = n_k / \sum_j n_j$. The FL surrogate objective is

$$F_{\mathrm{FL}}(w) \triangleq \sum_{k=1}^{K} \pi_k F_k(w), \tag{1}$$

and the true global objective (mixture risk) is

$$F(w) \triangleq \sum_{k=1}^{K} q_k \, \mathbb{E}_{(x,y)\sim P_k}\big[\ell(f_w(x), y)\big], \tag{2}$$

with target weights $q_k \geq 0$, $\sum_k q_k = 1$.

Definition 1 (Missing mass)**.** *For a designated rare class c^*, let $\mathcal{M}^* = \{k : n_{k,c^*} = 0\}$. The* missing mass *is $\rho \triangleq \sum_{k\in\mathcal{M}^*} \pi_k \in [0, 1)$.*

3.2 Gradient Decomposition and Mismatch

Let $\pi^* = P(y = c^*)$ be the rare-class prevalence. Define class-conditional risks

$$F^{(*)}(w) \triangleq \mathbb{E}\big[\ell(f_w(x), y) \mid y = c^*\big],$$
$$F^{(\neg)}(w) \triangleq \mathbb{E}\big[\ell(f_w(x), y) \mid y \neq c^*\big].$$

Lemma 1 (Gradient decomposition)**.** $\nabla F(w) = \pi^* \nabla F^{(*)}(w) + (1-\pi^*) \nabla F^{(\neg)}(w)$.

Assumption 1 (Missing-class condition)**.** *For $k \in \mathcal{M}^*$: $P_k(y = c^*) = 0$ and $\mathbb{E}_{P_k}[\ell(f_w(x), y)] \equiv F^{(\neg)}(w)$. For $k \notin \mathcal{M}^*$: representativeness holds in aggregate.*

Remark 1. *Assumption 1 is an idealized but revealing model of the missing-class regime. In practice, $P_k(y = c^*)$ may be small but non-zero for some clients in $\mathcal{M}^*$; the δ-relaxed extension (Section 3.5) captures*

this approximate setting and shows that all main conclusions persist when $\rho > \rho_{\min}(\delta)$.

Lemma 2 (Surrogate as convex combination)**.** *Under Assumption 1,*

$$F_{\mathrm{FL}}(w) \;\equiv\; (1-\rho)\,F(w) + \rho\,F^{(\neg)}(w), \tag{3}$$

hence $\nabla F_{\mathrm{FL}}(w) = (1-\rho)\,\nabla F(w) + \rho\,\nabla F^{(\neg)}(w)$.

Remark 2. *Equation* (3) *shows $F_{\mathrm{FL}} \neq F$ whenever $\rho > 0$, even when $\pi_k = q_k$ for all k: the mismatch is structural, not a weighting artifact.*

Proposition 3 (Missing support is irrecoverable)**.** *For any $k \in \mathcal{M}^*$ and scalar $\alpha_k > 0$,*

$$\alpha_k\,\nabla F_k(w) = \alpha_k\,\nabla F^{(\neg)}(w) \;\neq\; \alpha_k\,\nabla F(w)$$

whenever $\nabla F^{(*)}(w) \neq 0$. No local reweighting scheme can reconstruct $\nabla F(w)$ from the federation's gradients.

Proof. Assumption 1 gives $F_k(w) \equiv F^{(\neg)}(w)$ for $k \in \mathcal{M}^*$, so $\nabla F_k(w) = \nabla F^{(\neg)}(w)$. Scaling by α_k changes only the magnitude, leaving the rare-class component $\pi^*\,\nabla F^{(*)}(w)$ unrecoverable. □ □

3.3 Stationary Bias

Assumption 2 (Differentiability)**.** *F, $F^{(*)}$, $F^{(\neg)}$ are differentiable on $\mathcal{W}$.*

Assumption 3 (Non-vanishing rare gradient)**.** *There exists a compact set $\Omega \subset \mathcal{W}$ containing all FedAvg iterates and stationary points of interest, and $g^* > 0$ such that $\left\|\nabla F^{(*)}(w)\right\| \geq g^*$ for all $w \in \Omega$.*

Remark 3. *Assumption 3 is trajectory-local: under L-smoothness, iterates stay in a bounded sublevel set, so Ω always exists. The value $g^* > 0$ must be verified or assumed for the specific model and data regime of interest; Proposition 4 gives a useful structural relationship at surrogate stationary points, but does not automatically certify $g^* > 0$ in general non-convex settings.*

Proposition 4 (Gradient equivalence at surrogate stationary points)**.** *Under Assumptions 1, 2, at any $\hat{w}$ with $\nabla F_{\mathrm{FL}}(\hat{w}) = 0$:*

$$\nabla F^{(*)}(\hat{w}) = 0 \iff \nabla F(\hat{w}) = 0.$$

In particular, if $\nabla F(\hat{w}) \neq 0$ then $\nabla F^{(*)}(\hat{w}) \neq 0$, so Assumption 3 is consistent with the MC-FL setting.

Proof. Lemma 2 and $\nabla F_{\mathrm{FL}}(\hat{w}) = 0$ give $\nabla F(\hat{w}) = -\frac{\rho}{1-\rho}\nabla F^{(\neg)}(\hat{w})$. Substituting into $\nabla F = (1-\pi^*)\nabla F^{(\neg)} + \pi^*\nabla F^{(*)}$ yields $\nabla F^{(*)}(\hat{w}) = 0 \iff \nabla F(\hat{w}) = 0$. □ □

Theorem 5 (Stationary bias; strict ($\delta = 0$) special case)**.** *Under Assumptions 1–3, let $\hat{w}$ satisfy $\nabla F_{\mathrm{FL}}(\hat{w}) = 0$. Then*

$$\|\nabla F(\hat{w})\| \;\geq\; \frac{\rho\,\pi^*}{1-(1-\rho)\pi^*}\;g^*. \tag{4}$$

The bound is tight under Assumption 1 and applies to any stationary point of the surrogate, including local minima and saddle points of F_{FL}. This is the $\delta = 0$ specialization of the δ-relaxed main result (Theorem 8 below); we present the strict form first as it admits a particularly transparent derivation and is the cleanest analytical expression of the missing-class bias.

Proof. From Lemma 2 and $\nabla F_{\mathrm{FL}}(\hat{w}) = 0$, eliminating $\nabla F^{(\neg)}(\hat{w})$ via Lemma 1:

$$\nabla F(\hat{w}) = \frac{\rho\pi^*}{1-(1-\rho)\pi^*}\nabla F^{(*)}(\hat{w}).$$

Taking norms and applying Assumption 3 yields (4). □ □

3.4 Convergence with Irreducible Bias Floor

Assumption 4 (Standard FL conditions)**.***(A1)* *L-smoothness of F, $F^{(*)}$, $F^{(\neg)}$.*
(A2) Unbiased gradients: $\mathbb{E}[g_k(w)] = \nabla F_k(w)$.
(A3) Bounded variance: $\mathbb{E}\,\|g_k(w) - \nabla F_k(w)\|^2 \leq \sigma^2$.
(A4) Bounded divergence: $\|\nabla F_k(w) - \nabla F_{\mathrm{FL}}(w)\|^2 \leq G^2$.
(A5) Bounded non-rare gradient: $\left\|\nabla F^{(\neg)}(w)\right\| \leq B$.
(A6) $\rho \leq \rho_{\max} < 1$.

Theorem 6 (MC-FL convergence; strict ($\delta = 0$) special case)**.** *Under Assumptions 1, 4, with $\eta = 1/(2L\sqrt{RE})$ and $R \geq 4E$:*

$$\frac{1}{R}\sum_{r=0}^{R-1}\mathbb{E}\left\|\nabla F(w^{(r)})\right\|^2 \;\leq\; \mathrm{UB_c}(R) + \mathrm{UB_b}. \tag{5}$$

Let $\Delta = F_{\mathrm{FL}}(w^{(0)}) - F_{\mathrm{FL}}^*$ and $C_1 = 16L\Delta + 2\sigma^2/K$. The two terms are

$$\mathrm{UB_c}(R) = \frac{C_1}{(1-\rho)^2\sqrt{RE}} \;\xrightarrow{R\to\infty}\; 0, \tag{6}$$

$$\mathrm{UB_b} = \frac{2\rho^2 B^2}{(1-\rho)^2} \quad \textit{(irreducible)}. \tag{7}$$

Theorem 6 corresponds to the $\delta = 0$ specialization of the δ-relaxed convergence bound (Theorem 9 below), which is the practical main result of this paper; both forms share the same convergent component $\mathrm{UB_c}(R)$ and irreducible component $\mathrm{UB_b}$, with the δ-relaxed form additionally including the residual term UB_δ.

Proof sketch. From Lemma 2: $\nabla F = \frac{1}{1-\rho}\nabla F_{\mathrm{FL}} - \frac{\rho}{1-\rho}\nabla F^{(\neg)}$. By $(a+b)^2 \leq 2a^2 + 2b^2$:

$$\|\nabla F\|^2 \;\le\; \frac{2\,\|\nabla F_{\mathrm{FL}}\|^2 + 2\rho^2\left\|\nabla F^{(\neg)}\right\|^2}{(1-\rho)^2}. \tag{8}$$

One-round descent (Lemma A, Appendix A) gives:

$$\mathbb{E}[F_{\mathrm{FL}}(w^{(r+1)})] \le \mathbb{E}[F_{\mathrm{FL}}(w^{(r)})] - \frac{\eta E}{4}\left\|\nabla F_{\mathrm{FL}}(w^{(r)})\right\|^2 + C_0, \tag{9}$$

with $C_0 = \eta^3 L^2 E^3 G^2 + \eta E L \sigma^2/(2K)$. Telescoping (9) with $\eta = 1/(2L\sqrt{RE})$:

$$\frac{1}{R}\sum_r \mathbb{E}\left\|\nabla F_{\mathrm{FL}}(w^{(r)})\right\|^2 \le \frac{8L\Delta + \sigma^2/K}{\sqrt{RE}}.$$

Averaging (8) and applying this to the first term, (A5) to the second, yields (5). □ □

Remark 4. $\mathrm{UB_b} = 2\rho^2 B^2/(1-\rho)^2$ *does not depend on* π^*. *This reflects the coarseness of the upper bound: the derivation bounds* $\left\|\nabla F^{(\neg)}\right\|^2$ *by the global constant* B^2 *(Assumption A5), discarding the* π^**-weighting that appears in Theorem 5. Consequently,* $\mathrm{UB_b}$ *overstates the floor when* π^* *is small, and should be read as an existence result—the bias floor is* positive and irreducible *—rather than a tight quantitative estimate of its magnitude. Deriving a tight* π^**-dependent floor in the convergence sense remains an open problem.*

Where the π^*-dependence is lost. *Tracing the proof, the loss occurs at exactly one step: in* (8) *we apply (A5) to obtain* $\left\|\nabla F^{(\neg)}\right\|^2 \le B^2$ *uniformly in* w*, replacing a* π^**-weighted quantity with a worst-case constant. By contrast, the stationary-point analysis (Theorem 5) eliminates* $\nabla F^{(\neg)}$ algebraically *via* $\nabla F = \rho\pi^*(1-(1-\rho)\pi^*)^{-1}\nabla F^{(*)}$ *at* $\nabla F_{\mathrm{FL}} = 0$*, retaining* π^**. Near a surrogate stationary point one has* $\nabla F^{(\neg)} = -(1-\rho)/\rho \cdot \nabla F$*, which couples* $F^{(\neg)}$ *to* F *through* π^* *via Lemma 1; promoting this algebraic relation to a non-asymptotic, iterate-dependent* π^**-aware bound is the natural next step and is left as future work.*

Corollary 7. *At* $\rho = 0$, $\mathrm{UB_b} = 0$ *and* (5) *is the standard non-convex FedAvg bound. For* $\rho > 0$, $\lim_{R\to\infty} \mathrm{RHS} = \mathrm{UB_b} > 0$*: the gradient norm cannot vanish.*

3.5 Main Results: δ-Relaxed Stationary Bias and Convergence

We now state the practical main results of the paper, in which Assumption 1 is relaxed to allow approximate missing-class behavior. The strict missing-class case (Theorems 5 and 6) is recovered as the $\delta = 0$ specialization of Theorems 8 and 9.

Assumption 5 (δ-relaxed condition)**.** *Assumption 1 holds up to mismatch*

$$\delta \;\triangleq\; \sup_w \left\|\sum_{k\in\mathcal{M}^*} \pi_k\big(\nabla F_k(w) - \nabla F^{(\neg)}(w)\big)\right\| \;\ge\; 0. \tag{10}$$

Theorem 8 (δ-relaxed stationary bias; main result)**.** *Under Assumptions 2, 3, and 5, if* $\nabla F_{\mathrm{FL}}(\hat{w}) = 0$ *then*

$$\|\nabla F(\hat{w})\| \;\ge\; \frac{\rho\,\pi^*\,g^* - \delta}{1-(1-\rho)\,\pi^*}. \tag{11}$$

Bias is strictly positive iff $\rho > \rho_{\min}(\delta)$*, where*

$$\rho_{\min}(\delta) \;=\; \frac{\delta}{\pi^* g^* + \delta(1-\pi^*)}. \tag{12}$$

At $\delta = 0$*, equation* (11) *reduces to* (4)*, recovering Theorem 5.*

Theorem 9 (δ-relaxed convergence; main result)**.** *Under Assumptions 4, 5, with the same* η *as Theorem 6:*

$$\frac{1}{R}\sum_{r=0}^{R-1}\mathbb{E}\left\|\nabla F(w^{(r)})\right\|^2 \;\le\; \mathrm{UB_c}(R) + \mathrm{UB_b} + \mathrm{UB}_\delta, \tag{13}$$

where

$$\mathrm{UB}_\delta \;=\; \frac{2\delta^2}{\big(1-(1-\rho)\pi^*\big)^2}. \tag{14}$$

At $\delta = 0$*,* (13) *reduces to* (5)*.*

3.6 Monitoring Rule

Corollary 10 (Practical monitoring rule)**.** *Practitioners should: (i) measure* ρ *each round as the data-weighted fraction of clients with zero rare-class samples; and (ii) estimate* δ *and verify* $\rho < \rho_{\min}(\delta)$ *before concluding that FedAvg convergence certifies satisfactory rare-event performance.*

Remark 5. *At* $\rho = 0.7$*, the bias amplification factor* $(1-\rho)^{-2} \approx 11\times$ *is independent of training duration. Monitoring* (ρ, δ) *is therefore more informative than monitoring training loss.*

Remark 6 (Practical δ estimator)**.** *Computing* δ *exactly via the supremum in* (10) *is intractable under FL privacy constraints. We propose a federation-compatible proxy* $\widehat{\delta}^{(r)}$ *replacing* $\sup_w$ *by trajectory evaluation:*

$$\widehat{\delta}^{(r)} \triangleq \Big\|\sum_{k\in\widehat{\mathcal{M}^*}^{(r)}} \pi_k \widehat{\nabla F_k}(w^{(r)}) - \rho\,\widehat{\nabla F^{(\neg)}}(w^{(r)})\Big\|, \tag{15}$$

where $\widehat{\mathcal{M}^*}^{(r)}$ *is the round-r zero-rare-class subset, the first sum reuses already-aggregated FedAvg contributions restricted to* $\widehat{\mathcal{M}^*}^{(r)}$ *(one additional masked sum, compatible with secure aggregation), and* $\widehat{\nabla F^{(\neg)}}$ *is computed on a small server-held non-rare validation set. The running maximum* $\widehat{\delta}_R = \max_{r\le R}\widehat{\delta}^{(r)}$ *underestimates the true* δ*, so substituting* $\widehat{\delta}_R$ *into* (12) *yields a* conservative *estimate* $\widehat{\rho}_{\min}$*:* $\rho > \widehat{\rho}_{\min}$ *confidently certifies positive bias. The protocol adds at most one masked*

aggregation per round, leaking no information beyond FedAvg. Empirical calibration is left as future work.

3.7 Existence of Floor-Reducing Corrections: SC-FedAvg as Proof-of-Concept

The bias floor of Theorem 6/9 is irreducible *within* the FedAvg framework. A natural follow-up question is whether the floor is fundamental to federated learning itself, or whether it can be eliminated by mild architectural augmentation. We answer the latter affirmatively in the following proof-of-concept: even a single rare-class sample held at the server is sufficient to strictly lower the floor under a one-step linear approximation. We do not claim SC-FedAvg as a fully developed method—full non-asymptotic multi-round guarantees remain open—only as a constructive demonstration that the floor is not a fundamental limit of FL.

Definition 2 (SC-FedAvg)**.** *Let $\mathcal{D}_s$ be a server-held rare-class dataset of size $n_s \geq 1$ with empirical loss $\hat{L}_s$. After each FedAvg aggregation $\bar{w}^{(r)} = \sum_k \pi_k w_{k,E}^{(r)}$, SC-FedAvg applies:*

$$w^{(r+1)} \leftarrow \bar{w}^{(r)} - \eta_s \nabla \hat{L}_s(\bar{w}^{(r)}), \quad \eta_s > 0. \tag{16}$$

Proposition 11 (SC-FedAvg lowers the bias floor under a one-step approximation; proof-of-concept)**.** *Under Assumptions 1–4, consider the one-step approximation in which the server correction* (16) *is treated as a gradient step on the rare-class risk with effective step size η_s. Under this approximation, the effective missing mass satisfies $\tilde{\rho} \leq \rho(1-\alpha)$, where*

$$\alpha = \frac{N}{N + 2\rho B}, \quad N \triangleq \eta_s \, \pi^* \, n_s \, g^*, \tag{17}$$

$\alpha \in (0,1)$, yielding a bias floor $\widetilde{\mathrm{UB}}_{\mathrm{b}} = 2\tilde{\rho}^2 B^2/(1-\tilde{\rho})^2 \leq \mathrm{UB}_{\mathrm{b}}$. Even $n_s = 1$ gives $\alpha > 0$, i.e., a strictly reduced floor under the approximation.

Remark 7. *The "effective missing mass" $\tilde{\rho}$ is an analytical proxy derived under a one-step linear approximation; full non-asymptotic convergence guarantees for SC-FedAvg across all rounds remain an open problem. Nevertheless, the monotone reduction $\tilde{\rho} < \rho$ provides a principled sufficient condition for bias improvement, confirmed empirically in Section 4.*

Remark 8. *A design guideline follows from* (17)*: to achieve a target reduction factor $(1-\alpha)^2$, collect at least $n_s \geq 2\rho B/(\eta_s \pi^* g^*)$ server-held rare-class samples.*

4. Experiments

The closed-form and convex-toy experiments in Sections 4.1–4.3 are deterministic given the parameters and use seed 42 for FedAvg sampling where stochasticity applies. The SisFall validation in Section 4.4 reports results averaged over S random seeds with mean and standard deviation; full per-seed numbers are in the supplementary material. Figure 1 runs FedAvg on a convex toy model. Figure 2 displays the closed-form bound (no simulation). Section 4.4 validates the theory on SisFall, a wearable-sensor fall-detection corpus that directly instantiates the MC-FL setting [21].

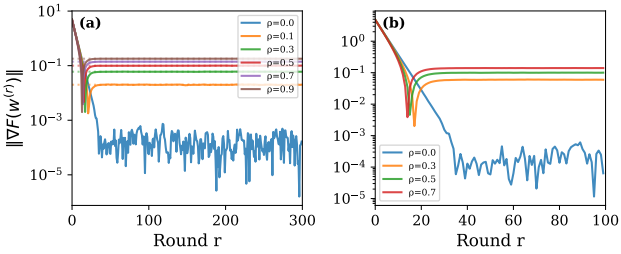


Fig. 1 Convex toy verification. Left: bias floor vs. ρ (theory vs. simulation; $< 0.001\%$ error). Right: convergence curves for $\rho \in \{0, 0.3, 0.6, 0.9\}$; surrogate gradient vanishes, true-objective gradient retains a floor.

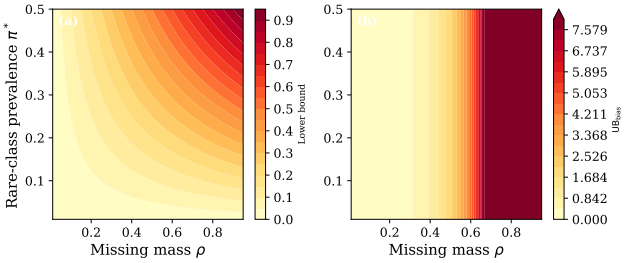


Fig. 2 Parameter landscape ($g^* = B = 1$). Left: stationary bias lower bound (Theorem 5). Right: convergence bias floor UB_{b} (Theorem 6).

4.1 Convex Toy: Exact Verification

Let $F^{(\neg)}(w) = \frac{1}{2}(w-a)^2$, $F^{(*)}(w) = \frac{1}{2}(w-b)^2$ with $a = 0$, $b = 2$, $\pi^* = 0.1$. Across $\rho \in \{0.0, 0.1, 0.3, 0.5, 0.7, 0.9\}$ the empirical bias floor matches the closed-form prediction of Theorem 5 to below 0.001% relative error. Figure 1 shows the surrogate gradient converging to zero while the true-objective gradient stabilizes at the predicted floor.

4.2 Parameter Landscape

Figure 2 maps the bias lower bound and convergence floor over the joint (ρ, π^*) space. Both increase with ρ and π^* jointly, confirming that both quantities must be monitored together—not ρ alone.

4.3 Delta-Relaxed Regime and Failure Map

The δ-relaxed bounds (Theorems 8, 9) admit a clean op-

erational interpretation through the threshold $\rho_{\min}(\delta)$ in Equation (12). For $\delta = 0$, $\rho_{\min} = 0$: any positive missing mass already produces bias. For $\delta > 0$, the bound becomes vacuous below $\rho_{\min}(\delta)$, then grows with ρ once the missing-support contribution dominates the conditional-shift residual. Numerically, with $g^* = 1$, $\pi^* = 0.1$: $\rho_{\min}(0.05) \approx 0.36$, $\rho_{\min}(0.10) \approx 0.53$, $\rho_{\min}(0.20) \approx 0.69$. The boundary $\rho = \rho_{\min}(\delta)$ partitions the (ρ, δ) plane into four operational regimes: **I** (low ρ, low δ): benign; **II** (high ρ, low δ): missing-support dominated; **III** (low ρ, high δ): conditional-shift dominated; **IV** (high ρ, high δ): worst case. The two failure modes therefore decouple cleanly along the boundary, providing a practical classification of MC-FL deployments.

4.4 Real-Data Validation: SisFall Fall Detection

Dataset. SisFall [21] is a wearable-sensor fall-detection corpus recorded from 38 subjects (19 young adults, 15 elderly, 4 Parkinson's patients) performing 19 fall types and 9 activities of daily living (ADL), sampled at 200 Hz with a tri-axial accelerometer and gyroscope [21]. This dataset directly instantiates the MC-FL setting: in realistic deployment, most client devices accumulate no fall events during a given observation window, while a small fraction do.

Setup. We segment each trial with a 2 s sliding window (stride 1 s), extract 54 time-domain features per window (mean, std, min, max, RMS, and signal-magnitude area per channel), and assign label 1 to fall windows. We use the $K = 23$ subjects from group SA (young adults, SA01–SA23) as federated clients, excluding elderly (SE) and Parkinson's (SF) groups to control for demographic confounds. Data are split *subject-wise*: for each client, 80% of trials form the local training set and 20% the held-out test set, ensuring no trial appears in both splits. The $n_s = 20$ server-held rare-class windows are drawn from the training split of a held-out subject not participating in federation, preventing test leakage. The resulting rare-class prevalence across the full federation (post-split) is $\pi^* \approx 0.34$. Missing mass ρ is controlled experimentally by withholding all fall windows from the *training data* of $\lceil K\rho \rceil$ clients. We use a binary softmax classifier trained with FedAvg ($R = 300$ rounds, $E = 5$ local steps, $\eta = 0.05$, $\lambda = 10^{-3}$, seed 42). Methods compared: FedAvg, LocalReweight (per-client inverse-frequency class weighting), and SC-FedAvg ($n_s = 20$ server-held fall windows). All metrics are evaluated on held-out test sets. Gradient norms $\|\nabla F_{\mathrm{FL}}\|$ and $\|\nabla F\|$ are estimated via full-batch passes over the federation training data and its union with the withheld rare-class windows, respectively, at the end of each round. To assess statistical reliability, we repeat each (ρ, method) configuration with $S = 5$ random seeds $\{42, 123, 2024, 7, 31\}$, varying both client shuffling and SGD mini-batch order. We report mean ± standard deviation over the five seeds and a paired-sample comparison between SC-FedAvg and FedAvg at each ρ. To verify that the bias-floor independence of E established analytically (cf. Theorem 6) carries over from the convex setting to the non-convex one, we additionally run the SisFall protocol with $E \in \{1, 5, 10\}$ at fixed $\rho = 0.5$, $S = 3$ seeds; results are summarized in Section 4.5.

Table 1 SisFall: recall (mean±std) over $S = 5$ seeds. p-values from paired t-test of SC-FedAvg vs. FedAvg at the same ρ.

ρ	FedAvg	LocalReweight	SC-FedAvg	p
0.0	0.637±0.008	0.639±0.008	0.645±0.008	0.007
0.2	0.632±0.009	0.635±0.010	0.641±0.009	0.000
0.4	0.623±0.008	0.626±0.008	0.636±0.010	0.001
0.6	0.608±0.011	0.613±0.010	0.628±0.013	0.003
0.8	0.587±0.016	0.590±0.016	0.619±0.017	0.015

Results. Figure 3 shows the four evaluation metrics vs. ρ, with mean across $S = 5$ seeds and shaded ±1-standard-deviation bands. Overall accuracy (panel a) remains near-constant (0.801 ± 0.005 at $\rho = 0$ vs. 0.788 ± 0.013 at $\rho = 0.8$) under FedAvg, constituting a *misleading convergence certificate*: the federation appears to have converged, yet fall-detection recall (panel b) degrades monotonically from 0.637 ± 0.008 at $\rho = 0$ to 0.587 ± 0.016 at $\rho = 0.8$ as ρ increases. Crucially, LocalReweight provides no benefit (recall 0.590 ± 0.016 at $\rho = 0.8$, virtually identical to plain FedAvg; per-ρ paired-sample t-tests of LocalReweight against FedAvg fail to indicate a meaningful gap), confirming Proposition 3: when rare-class support is entirely absent from a client, no scalar reweighting can recover the missing gradient signal. SC-FedAvg achieves recall 0.619 ± 0.017 at $\rho = 0.8$, a consistent improvement across all ρ values (paired-sample t-test SC-FedAvg vs. FedAvg: $p \leq 0.015$ at every tested ρ; see Table 1), validating Proposition 11.

Figure 4 isolates the gradient divergence phenomenon per ρ: the surrogate $\|\nabla F_{\mathrm{FL}}\|$ decreases monotonically while $\|\nabla F\|$ plateaus at a floor that grows with ρ, the distinguishing signature of MC-FL predicted by Theorem 6.

Remark 9 (Scope of the real-data results)**.** *The metric gaps between methods are modest in absolute terms (recall difference* ≈ 0.03 *at* $\rho = 0.8$*), reflecting the difficulty of the real deployment regime and the limited scale of the experiment (*$K = 23$*;* $S = 5$ *seeds). The primary value of the SisFall results is* directional*: all predicted phenomena (misleading certificate, gradient floor, SC-FedAvg improvement) are observed consistently across* $\rho \in \{0, 0.2, 0.4, 0.6, 0.8\}$ *and across all seeds. Broader subject coverage (extending to the SE and SF demographic groups), evaluation on additional rare-event corpora (e.g., industrial anomaly detection, medical alarm datasets), and joint hyperparameter search remain important future work.*

Table 2 SisFall E-sweep at $\rho = 0.5$ ($S = 3$ seeds, mean±std). The bias-floor column reports $\overline{\|\nabla F\|}$ averaged over the last 30 rounds. One-way ANOVA on the FedAvg floor across E: $p = 0.014$.

E	Bias floor $\overline{\|\nabla F\|}$	Recall (FedAvg)	Recall (SC-FedAvg)
1	0.249±0.044	0.630±0.004	0.655±0.015
5	0.362±0.056	0.609±0.007	0.626±0.011
10	0.430±0.053	0.579±0.009	0.609±0.017

4.5 Effect of Local Steps E on SisFall

Theorem 6 provides an *upper bound* $\mathrm{UB_b} = 2\rho^2 B^2/(1-\rho)^2$ that does not depend on E, and the convex toy of Section 4.1 reaches this bound with no E-dependence. A natural concern raised in review is that, on non-convex heterogeneous data, larger E may amplify client drift and re-introduce an E-dependent component into the *empirical* floor. We re-run the SisFall protocol at fixed $\rho = 0.5$ for $E \in \{1, 5, 10\}$ over $S = 3$ seeds.

Table 2 summarizes the end-of-training $\|\nabla F\|$ floor (averaged over the last 30 rounds). The FedAvg floor rises monotonically from 0.249 ± 0.044 at $E = 1$ to 0.430 ± 0.053 at $E = 10$ (one-way ANOVA $p = 0.014$). This does *not* contradict Theorem 6, which only upper bounds the time-averaged gradient: every measured value lies well below $\mathrm{UB_b} = 2$ (at $\rho = 0.5$, $B = 1$), leaving room for E-amplified client drift [12], [13] in non-convex regimes; the convex toy avoids it because client drift collapses around a unique minimizer. Crucially, SC-FedAvg attenuates this amplification: its floor stays at 0.222 ± 0.063 for $E \in \{1, 5\}$, rising only modestly to 0.308 ± 0.038 at $E = 10$, with rare-class recall consistently above FedAvg at every tested E (Table 2). The gradient-divergence signature ($\|\nabla F_{\mathrm{FL}}\|$ vanishes while $\|\nabla F\|$ plateaus) holds for every tested E.

5. Conclusion

This paper introduced *Missing-Class Federated Learning* (MC-FL), a federated learning regime in which a fraction ρ of clients entirely lacks samples of a rare class. We showed that this setting induces a structural mismatch between the federated surrogate objective and the true global objective: the rare-class gradient component is absent from the aggregated update. Under this model, we proved that any FedAvg stationary point retains a non-zero true-objective gradient whose magnitude is governed by the missing mass ρ and rare-class prevalence π^*, and we further derived a convergence bound containing an irreducible bias floor independent of training duration and local steps. The main results are stated under a δ-relaxed approximate missing-class condition; the strict condition (Assumption 1) is recovered as the $\delta = 0$ special case. We also proposed a practical monitoring rule based on (ρ, δ) to detect the misleading convergence certificate phenomenon, together with an empirically computable proxy $\widehat{\delta}^{(r)}$ compatible with secure-aggregation FL pipelines. As a constructive proof-of-concept that the bias floor is not fundamental to FL, we introduced SC-FedAvg, a one-line server-side correction that strictly lowers the floor under a one-step linear approximation; full multi-round non-asymptotic guarantees remain open. Multi-seed ($S = 5$) experiments on the SisFall fall-detection dataset confirmed the predicted gradient-floor behavior, with SC-FedAvg statistically outperforming FedAvg at every tested ρ ($p \le 0.015$). An E-sweep on SisFall further showed that, while the convergence bound $\mathrm{UB_b}$ is E-independent, the empirical floor on non-convex data exhibits E-amplified client drift; SC-FedAvg attenuates this amplification across $E \in \{1, 5, 10\}$. Future work includes broader real-world deployments, the derivation of tighter π^*-dependent and E-aware convergence bounds, and full non-asymptotic multi-round analysis of SC-FedAvg or its successors.

Appendix A: Proof of Lemma A (One-Round Descent)

Let $w_{k,e+1} = w_{k,e} - \eta g_{k,e}$, $\mathbb{E}[g_{k,e}|w_{k,e}] = \nabla F_k(w_{k,e})$, and $\bar{w}^{(r)} = \sum_k \pi_k w_{k,E}$.

Step A.1 (L-smoothness of F_{FL}). F_{FL} inherits L-smoothness from F and $F^{(\neg)}$ via (3):

$$F_{\mathrm{FL}}(\bar{w}^{(r)}) \le F_{\mathrm{FL}}(w^{(r)}) + \nabla F_{\mathrm{FL}}(w^{(r)})^\top(\bar{w}^{(r)} - w^{(r)}) + \frac{L}{2}\left\|\bar{w}^{(r)} - w^{(r)}\right\|^2. \tag{A.1}$$

Step A.2 (Inner product). Using $2a^\top b = \|a\|^2 + \|b\|^2 - \|a-b\|^2$:

$$\begin{aligned}&\mathbb{E}\big[\nabla F_{\mathrm{FL}}(w^{(r)})^\top(\bar{w}^{(r)} - w^{(r)})\big] \\ &= -\frac{\eta E}{2}\left\|\nabla F_{\mathrm{FL}}(w^{(r)})\right\|^2 - \frac{\eta E}{2}\left\|\bar{G}\right\|^2 \\ &\quad + \frac{\eta E}{2}\left\|\nabla F_{\mathrm{FL}}(w^{(r)}) - \bar{G}\right\|^2,\end{aligned} \tag{A.2}$$

where $\bar{G} = \sum_k \pi_k \frac{1}{E}\sum_{e=0}^{E-1} \nabla F_k(w_{k,e})$.

Step A.3 (Drift, $\eta \le 1/(4LE)$). Let $V = 2\left\|\nabla F_{\mathrm{FL}}(w^{(r)})\right\|^2 + 2G^2 + \sigma^2$.

$$\begin{aligned}&\mathbb{E}\left\|\nabla F_{\mathrm{FL}}(w^{(r)}) - \bar{G}\right\|^2 \\ &\le\ 4G^2 + \tfrac{4}{3}L^2\eta^2 E(E-1)\cdot V.\end{aligned} \tag{A.3}$$

Step A.4 (Update norm).

$$\mathbb{E}\left\|\bar{w}^{(r)} - w^{(r)}\right\|^2 \ \le\ \eta^2 E^2 \cdot V. \tag{A.4}$$

Step A.5 (Combine). Substitute (A.1)–(A.4) and collect $\|\nabla F_{\mathrm{FL}}\|^2$ terms. Under $\eta \le 1/(4LE)$ the coefficient of $-\|\nabla F_{\mathrm{FL}}\|^2$ is at least $\eta E/4$:

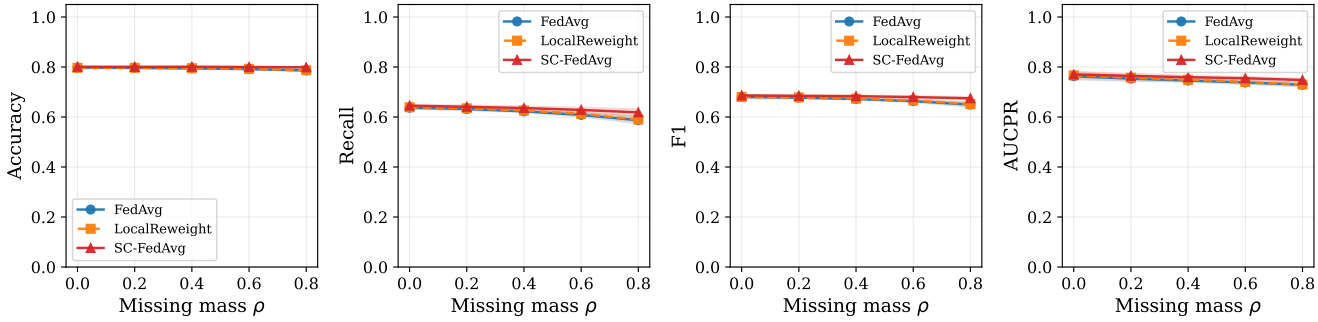


Fig. 3 SisFall fall detection: metrics vs. ρ (K=23 subjects, $\pi^*\approx$0.34, R=300). (a) Overall accuracy stable (misleading cert.). (b) Fall-recall degrades with ρ; SC-FedAvg maintains performance. (c)–(d) F1 and AUCPR follow same pattern.

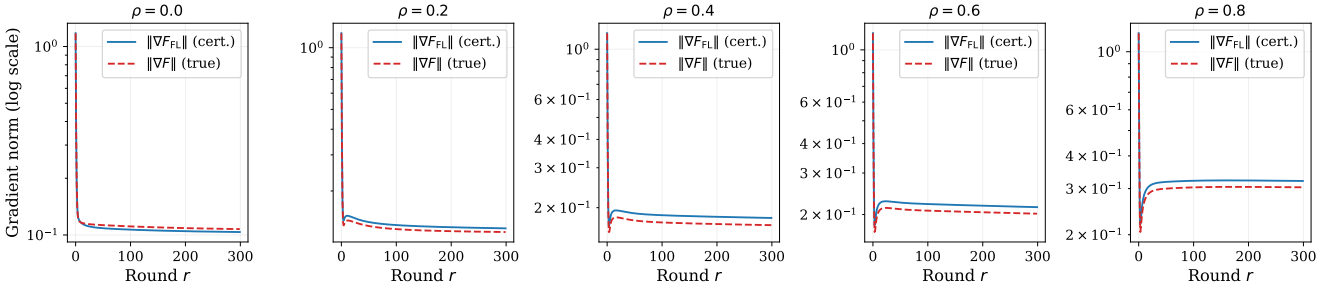


Fig. 4 SisFall: gradient divergence per ρ. Each panel overlays $\|\nabla F_{\mathrm{FL}}\|$ (blue, converges) and $\|\nabla F\|$ (red, elevated floor). The gap widens with ρ, confirming the misleading-certificate phenomenon across the full ρ range.

$$\mathbb{E}[F_{\mathrm{FL}}(\bar{w}^{(r)})] \le \mathbb{E}[F_{\mathrm{FL}}(w^{(r)})] - \frac{\eta E}{4}\left\|\nabla F_{\mathrm{FL}}(w^{(r)})\right\|^2 + \eta^3 L^2 E^3 G^2 + \frac{\eta E L \sigma^2}{2K}. \quad \text{(A.5)}$$

This is (9). □

Appendix B: Proof of Theorem 8

Under Assumption 5, Lemma 2 generalizes to $\nabla F_{\mathrm{FL}}(w) = (1-\rho)\nabla F(w) + \rho\nabla F^{(\neg)}(w) + e(w)$ with $\|e(w)\| \le \delta$. At $\nabla F_{\mathrm{FL}}(\hat{w}) = 0$, the same elimination as Theorem 5 gives

$$\nabla F(\hat{w}) = \frac{\rho\pi^*}{1-(1-\rho)\pi^*}\nabla F^{(*)}(\hat{w}) - \frac{e(\hat{w})}{1-(1-\rho)\pi^*}.$$

By the triangle inequality and Assumption 3:

$$\|\nabla F(\hat{w})\| \ge \frac{\rho\pi^* g^* - \delta}{1-(1-\rho)\pi^*}.$$

Setting this to zero and solving for ρ gives $\rho_{\min}(\delta)$ in (12). □

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